ATLAS — Voice Bridge Module

Technical summary — final year project

Models Used

- Speech-to-Text: OpenAI Whisper (small) — multilingual, auto-detects spoken language
- Translation: Meta NLLB-200-distilled-600M — supports 200 languages via FLORES-200 codes
- Text-to-Speech: Google Text-to-Speech (gTTS) — online, ~100+ languages
- Text language ID: fastText lid.176.bin — offline, 176 languages

Datasets

- No custom training — all models are pretrained by OpenAI / Meta on public corpora
- Whisper: 680,000 hours of multilingual audio
- NLLB-200: FLORES-200 parallel translation dataset
- fastText: Wikipedia + web text (176 languages)
- Romanized-language word lists (Hindi, Tamil, Bengali, etc.) are hand-curated, not a formal dataset

Custom Logic Built

- Script detection: Unicode character-range check identifies native script instantly (100% reliable)
- Romanized detection: hand-built dictionaries for 9 Indian languages, scored by word overlap
- Transliteration: Google Input Tools API (online) with offline ITRANS fallback (indic_transliteration)
- 4-tier pipeline: script check → romanized dictionary → fastText → langdetect fallback

Key Libraries

- openai-whisper, transformers (Hugging Face), torch
- gtts, fasttext-wheel, langdetect, indic_transliteration
- requests (Google Input Tools calls), fastapi (backend)

Architecture / Flow

- Voice input: browser mic → /api/voice/converse → Whisper → NLLB → gTTS → response + audio URL
- Text input: typed text → /api/voice/converse-text → 4-tier detection → NLLB → gTTS → response
- Audio served via FastAPI StaticFiles mount at /outputs, played inline with HTML5

Language Coverage

- Translation: ~100+ languages configured (subset of NLLB-200's 200)
- Speech output: ~85+ of those languages (limited by gTTS support)
- Auto-detection: 21 native scripts + 9 major Indian languages in romanized form